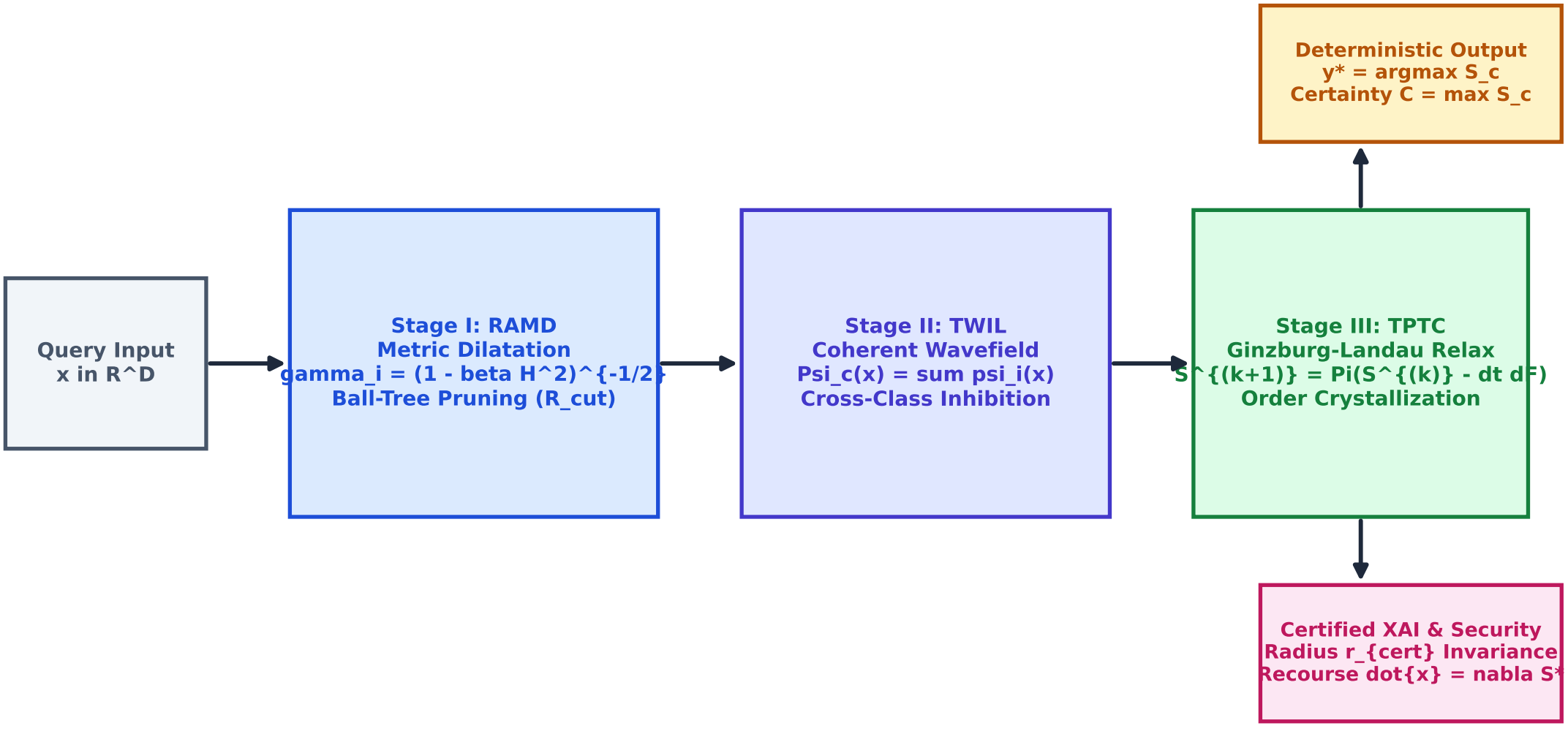TRTL End-to-End Inference Pipeline: From Metric Warping to Thermodynamic Crystallization